

Taste and smell

Detecting flavors depends as much on the nose as on the taste buds. The nose picks up external odors from food (see pp.78–79), but this is increased significantly by food-particle odors carried up into the nasal cavity by expired air from the lungs (retronasal olfaction). Some smell receptors have also been found in the taste buds. The brain combines the information from the nose and taste buds to perceive all the different flavors in the food. These are not the only sensations that contribute to the taste experience—the somatosensory cortex detects the texture and temperature of food, adding context to the flavor.

Signals travel to secondary taste area, located in orbitofrontal cortex

ORBITOFRONTAL CORTEX

Signals sent to primary taste area, located in insula

Signals travel to tongue area of somatosensory cortex

SOMATOSENSORY CORTEX

Signals from olfactory cortex sent to orbitofrontal cortex

Olfactory cortex

OLFACTORY BULB

NASAL CAVITY

THALAMUS

AMYGDALA

Amygdala assigns positive or negative values to taste and smell

MEDULLA

Smell from food particles that have been swallowed are sent for processing by olfactory bulb

The taste pathway

Information from the taste buds travels to the brain via cranial nerves in the jaw and throat. Impulses travel up the brain stem to the thalamus and are forwarded to the taste regions of the frontal cortex and insula, a fold of cortex deep in the brain.

Food particle

Trigeminal and glossopharyngeal cranial nerves carry signals to medulla in brain stem

Expired air from lungs pushes food particles from mouth into nasal cavity

WHY DON'T BABIES LIKE BITTER FOODS?

Babies have many more taste buds than adults so they taste bitter foods more intensely. They instinctively refuse foods that aren't as sweet or fatty as breast milk.

KEY

- Taste signals
- Retronasal smell
- Expired air

THE AVERAGE ADULT HAS BETWEEN 2,000 AND 8,000 TASTE BUDS

